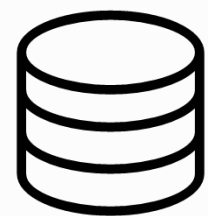


Learning Procedure

1. Offline

Offline
Dataset (Opt.)



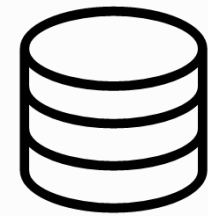
Offline
Pretrain L_{IL}

π_{VLA}

2. Online

Server Learner

Replay Buffer



Online Finetune
 $L_{IL} + L_{RL}$

π_{VLA} π_{edit} Q

Update per step/
episode/large batch

Sync/Async (Opt.)

Actor

(s, a, r, s')

Environment
Sensors
Controller
Success
Detector

Online Rollout
Human Intervention

Policy Architecture

Action Proposals

Edit Actions

π_{VLA}

VLA Model

States

Prompt Images Proprios

$a_{t:t+H}^1$
 $a_{t:t+H}^2$
...
 $a_{t:t+H}^N$

π_{edit}

Edit
Actor

States

$\hat{a}_{t:t+C}^1$
 $\hat{a}_{t:t+C}^2$
...
 $\hat{a}_{t:t+C}^N$

$\operatorname{argmax}_a Q(s, a_{t:t+C})$

$[\overline{a_{t:t+C} \times N} \quad \overline{a_{t:t+C} + \hat{a}_{t:t+C} \times N}] \rightarrow a_{t:t+C}^*$